Lecture note, Electrodynamics II: 7120

Wednesday, April 19, 2017

Lecture note: Derivation of the constitutive relations $\mathbf{D}(\mathbf{E},\mathbf{H})$ and $\mathbf{B}(\mathbf{E},\mathbf{H})$ in the moving media. (This note first appeared as a solution to a homework problem.)

(a) Check by explicit calculations that field tensors

$$F^{\mu\nu} = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{pmatrix} \tag{1}$$

and

$$H^{\mu\nu} = \begin{pmatrix} 0 & -cD_x & -cD_y & -cD_z \\ cD_x & 0 & -H_z & H_y \\ cD_y & H_z & 0 & -H_x \\ cD_z & -H_y & H_x & 0 \end{pmatrix} \tag{2}$$

describe all Maxwell's equations, with no sources, in a uniform and isotropic medium (characterized by ϵ and μ) via

$$\frac{\partial F_{\lambda\mu}}{\partial x^\nu} + \frac{\partial F_{\mu\nu}}{\partial x^\lambda} + \frac{\partial F_{\nu\lambda}}{\partial x^\mu} = 0 \tag{3}$$

and

$$\frac{\partial H^{\lambda\mu}}{\partial x^\mu} = 0 \tag{4}$$

(b) How the last equation changes in the presence of free charge and current densities ρ_{free} and $\mathbf{J}_{\text{free}}$? Write it in a 4-vector form.

(c) Given that in a rest-frame of the medium the constitutive relations are $\mathbf{D} = \epsilon\mathbf{E}$ and $\mathbf{B} = \mu\mathbf{H}$, establish proportionality relations between the following combinations $H^{\mu\nu}u_\nu$ and $F^{\mu\nu}u_\nu$, and between $F_{\lambda\mu}u_\nu + F_{\mu\nu}u_\lambda + F_{\nu\lambda}u_\mu$ and $H_{\lambda\mu}u_\nu + H_{\mu\nu}u_\lambda + H_{\nu\lambda}u_\mu$. Here $u_\nu = \gamma(c, -\mathbf{v})$ is a 4-velocity and $\gamma = 1/\sqrt{1 - v^2/c^2}$. Remember that in the rest frame $u_\nu = (c, 0)$.

(d) Use results from part (c) to establish the constitutive relations between vectors $\mathbf{D}, \mathbf{E}, \mathbf{B}$, and $\mathbf{H}$, similar to how it was done in the class.

Derivation of these results follows below:

P1

$$\frac{\partial F_{\lambda\mu}}{\partial x^{\nu}} + \frac{\partial F_{\mu\nu}}{\partial x^{\lambda}} + \frac{\partial F_{\nu\lambda}}{\partial x^{\mu}} = 0$$ All 3 indices must be different otherwise LHS = 0 trivially: for example, let $\mu = \nu \Rightarrow$

$$\frac{\partial F_{\lambda\mu}}{\partial x^{\mu}} + \frac{\partial F_{\mu\mu}^{=0}}{\partial x^{\lambda}} + \frac{\partial F_{\mu\lambda}}{\partial x^{\mu}} =$$

$$= \frac{\partial}{\partial x^{\mu}}\left(F_{\lambda\mu} + F_{\mu\lambda}\right) = 0.$$

So, let $\lambda = 0, \mu = 1, \nu = 2$:

$$\frac{\partial F_{01}}{\partial x^{2}} + \frac{\partial F_{12}}{\partial x^{0}} + \frac{\partial F_{20}}{\partial x^{1}} = 0$$

$$\frac{\partial (E_x/c)}{\partial y} + \frac{\partial(-B_z)}{\partial(ct)} + \frac{\partial}{\partial x}\left(-\frac{E_y}{c}\right) = 0$$

$$\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} + \frac{\partial B_z}{\partial t} = 0$$

$$\left(\vec{\nabla} \times \vec{E}\right)_z = -\frac{\partial B_z}{\partial t}$$

$\Rightarrow$ this eqn. gives us $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

$$\frac{\partial H^{\lambda\mu}}{\partial x^{\mu}} = 0$$

$\lambda = 0$: $\quad -c\,\vec{\nabla} \cdot \vec{D} = 0$

$\lambda = 1$: $\quad \frac{\partial (cD_x)}{c\,\partial t} + \frac{\partial}{\partial y}(-H_z) + \frac{\partial}{\partial z} H_y = 0$

$$\frac{\partial}{\partial y} H_z - \frac{\partial H_y}{\partial z} = \frac{\partial D_x}{\partial t} \Rightarrow \left(\vec{\nabla} \times \vec{H}\right)_x = \left(\frac{\partial \vec{D}}{\partial t}\right)_x$$

(b) $J^\alpha = (c\rho, \vec{J})$

Expect $\dfrac{\partial H^{\lambda\mu}}{\partial x^\mu} = a J^\lambda$ $\Rightarrow$ need to find a

$\lambda = 0$: $-c\vec{\nabla}\cdot\vec{D} = a J^0 = a c \rho$

$\lambda = 1$:

$$\frac{c\,\partial D_x}{c\,\partial t} - \left(\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z}\right) = a J^1 = a J_x$$

$$\left(\vec{\nabla}\times\vec{H}\right)_x = \frac{\partial D_x}{\partial t} - a J_x = \left(\frac{\partial \vec{D}}{\partial t} - a\vec{J}\right)_x$$

Must be: $\left.\begin{aligned} \vec{\nabla}\cdot\vec{D} &= \rho \\ \vec{\nabla}\times\vec{H} &= \vec{J} + \frac{\partial \vec{D}}{\partial t} \end{aligned}\right\} \Rightarrow a = -1.$

$$\Rightarrow \frac{\partial H^{\lambda\mu}}{\partial x^\mu} = -J^\lambda$$

(c) $u^\mu = \gamma(1, \vec{v}/c)$, $\gamma = \dfrac{1}{\sqrt{1 - v^2/c^2}}$

In the rest frame: $F^{\mu\nu}u_\nu = \begin{cases} \frac{1}{c}E_\mu & \mu = 1,2,3 \quad x\,y\,z \\ 0 & \mu = 0 \end{cases}$

$u^\mu = (1, 0)$

$$H^{\mu\nu}u_\nu = \begin{cases} c D_\mu & \mu = 1,2,3 \\ 0 & \mu = 0 \end{cases}$$

$$\vec{D} = \varepsilon\vec{E} \Rightarrow c D_\mu = \varepsilon c^2\left(\frac{1}{c}E_\mu\right)$$

Thus $H^{\mu\nu}u_\nu = c^2\varepsilon F^{\mu\nu}u_\nu$ (1)

$F_{\lambda\mu} u_\nu + F_{\mu\nu} u_\lambda + F_{\nu\lambda} u_\mu$ must have all indices different

$\lambda = 0, \mu = 1, \nu = 2$:

$$F_{01} \underset{0}{\underset{\parallel}{u_2}} + F_{12} \underset{1}{\underset{\parallel}{u_0}} + F_{20} \underset{0}{\underset{\parallel}{u_1}} = -B_z$$

$$\left(H_{\lambda\mu} u_\nu + H_{\mu\nu} u_\lambda + H_{\nu\lambda} u_\mu\right)_{\substack{\lambda=0 \\ \mu=1 \\ \nu=2}} = -H_z$$

$\vec{B} = \mu \vec{H} \Rightarrow$ the relation is:

$$(2) \quad \left(F_{\lambda\mu} u_\nu + F_{\mu\nu} u_\lambda + F_{\nu\lambda} u_\mu\right) = \mu \left(H_{\lambda\mu} u_\nu + H_{\mu\nu} u_\lambda + H_{\nu\lambda} u_\mu\right)$$

(d) Now apply eqns (1) and (2) in a moving frame, where $u_\nu = \gamma(1, -\vec{v}/c)$

<u>Eq. 2</u> Here $\underset{\frac{E_x}{c}}{\underset{\parallel}{F_{01}}} \underset{-\gamma \frac{v_y}{c}}{\underset{\parallel}{u_2}} + \underset{-B_z}{\underset{\parallel}{F_{12}}} \underset{\gamma}{\underset{\parallel}{u_0}} + \underset{-\frac{E_y}{c}}{\underset{\parallel}{F_{20}}} \underset{-\gamma \frac{v_x}{c}}{\underset{\parallel}{u_1}} =$

$$= \gamma\left(-B_z + \frac{1}{c^2}(v_x E_y - v_y E_x)\right) = \gamma\left(-B_z + \frac{1}{c^2}(\vec{v} \times \vec{E})_z\right)$$

$$H_{01} u_2 + H_{12} u_0 + H_{20} u_1 =$$

$$= \gamma\left(-H_z + cD_x(-v_y/c) + (-cD_y)(-v_x/c)\right) =$$

$$= \gamma\left(-H_z + v_x D_y - v_y D_x\right) = \gamma\left(-H_z + (\vec{v} \times \vec{D})_z\right)$$

So we have $-\gamma\left(\vec{B} - \frac{1}{c^2}\vec{v}\times\vec{E}\right)_z = \mu(-\gamma)\left(\vec{H} - \vec{v}\times\vec{D}\right)_z$

$$\boxed{\vec{B} - \frac{1}{c^2}\vec{v}\times\vec{E} = \mu\left(\vec{H} - \vec{v}\times\vec{D}\right).}$$

Eq 1

$$H^{\mu\nu}u_\nu\Big|_{\mu=0} = -c\vec{D}\cdot\left(-\frac{\gamma}{c}\vec{v}\right) = \gamma\,\vec{D}\cdot\vec{v}$$

$$H^{\mu\nu}u_\nu\Big|_{\mu=1} = cD_x\gamma - H_z\left(-\frac{\gamma}{c}v_y\right) + H_y\left(-\frac{\gamma}{c}v_z\right) =$$

$$= \gamma\left(cD_x + \frac{1}{c}\left(\vec{v}\times\vec{H}\right)_z\right)$$

$$F^{\mu\nu}u_\nu\Big|_{\mu=0} = -\frac{1}{c}\vec{E}\cdot\left(-\frac{\gamma}{c}\vec{v}\right) = \frac{\gamma}{c^2}\vec{E}\cdot\vec{v}$$

$$F^{\mu\nu}u_\nu\Big|_{\mu=1} = \frac{1}{c}E_x\gamma + \frac{\gamma}{c}B_z v_y - \frac{\gamma}{c}B_y v_z =$$

$$= \frac{\gamma}{c}\left(E_x + \left(\vec{v}\times\vec{B}\right)_x\right)$$

From $H^{\mu\nu}u_\nu = c^2\varepsilon F^{\mu\nu}u_\nu$ follows:

$$\gamma\,\vec{D}\cdot\vec{v} = c^2\varepsilon\,\frac{\gamma}{c^2}\vec{E}\cdot\vec{v} = \varepsilon\gamma\,\vec{E}\cdot\vec{v}$$

$$\Rightarrow \vec{D}\cdot\vec{v} = \varepsilon\,\vec{E}\cdot\vec{v} \quad \text{i.e. } \vec{D}_{\parallel} = \varepsilon\,\vec{E}_{\parallel}$$ along axis of motion.

$$\gamma\left(c\vec{D} + \frac{1}{c}\vec{v}\times\vec{H}\right) = c^2\varepsilon\,\frac{\gamma}{c}\left(\vec{E} + \vec{v}\times\vec{B}\right)$$

$$\boxed{\vec{D} + \frac{1}{c^2}\vec{v}\times\vec{H} = \varepsilon\left(\vec{E} + \vec{v}\times\vec{B}\right)}$$